Supplementary Materials

Actively Sold Books

ISBN	Title
9780349114033	Four Blondes
9780241003008	Very Hungry Caterpillar, The
9780552998482	Chocolat: (Chocolat 1)
9781841150437	Bad Blood: A Memoir
9780006647553	Mog’s Bad Thing
9780006531203	In the Heart of the Sea: The Epic True Story That Inspired ‘Moby Dick’
9780552145954	Between Extremes
9780091816971	Who Moved My Cheese
9780140259506	My East End: Memories of Life in Cockney London
9780749395698	Mr Nice
9780752844299	Arthur: The Seeing Stone: Book 1: Arthur
9780552998727	Marrying The Mistress
9780006514091	Glass Palace, The
9780006512134	Man and Boy
9780091867775	Round Ireland with a Fridge
9780140276336	White Teeth
9780440864554	Jacqueline Wilson's Superstars
9780349112763	Fortune's Rocks
9780099771517	Memoirs of a Geisha: The Literary Sensation and Runaway Bestseller
9780749397548	Captain Corelli's Mandolin: AS SEEN ON BBC BETWEEN THE COVERS
9781841462400	KS3 Science Revision Guide - Foundation CGP KS3 Revision Guides
9780349113609	Last Time They Met, The
9780552998444	Best a Man Can Get, The
9780719559792	Now We Are Sixty: 20th Anniversary Edition
9780261103252	Lord of the Rings, The
9780340766057	McCarthy's Bar: A Journey of Discovery in Ireland
9780099285823	Experience
9780099428558	Howard Marks' Book Of Dope Stories

9781841460406	KS3 Maths Revision Guide - Foundation CGP KS3 Revision Guides
9780552997034	Down Under
9780340696767	Nathaniel's Nutmeg: How One Man's Courage Changed the Course of History
9780140275421	Other Side of the Dale, The
9781841461502	KS2 English Study Book - Ages 7-11: CGP KS2 English
9780593048153	Universe In A Nutshell, The
9781841462301	KS3 Science Revision Guide - Higher CGP KS3 Revision Guides
9780099286578	Elizabeth
9781841462509	KS2 Science Study Book: CGP KS2 Science
9780744523232	We're Going on a Bear Hunt
9780140281293	Over Hill and Dale
9780140285215	English Passengers
9780722532935	Alchemist, The
9781841460307	KS3 Maths Revision Guide - Higher CGP KS3 Revision Guides
9780140276619	Consolations

Table S1: List of 61 books that remain actively sold in the market with recorded sales beyond 1 July 2024. ISBNs and raw titles are shown as collected from source data.

Weekly Data

The Alchemist

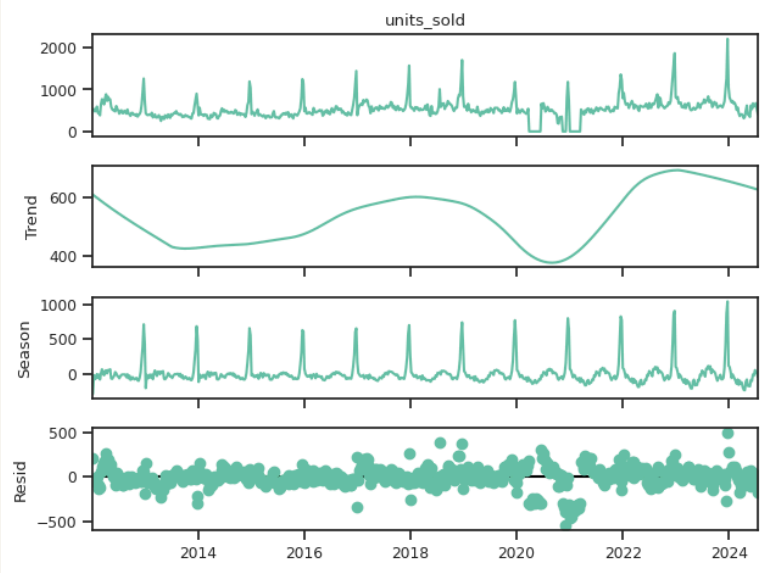


Figure S1. STL decomposition of weekly sales for The Alchemist (trend=157, seasonal=25, period=52). Seasonal-Trend decomposition showing moderate trend strength (0.49) with gradual growth and COVID-19 disruption (2020-2021), strong seasonal component (0.68) with regular annual spikes, and residuals containing irregular changes with occasional extreme deviations (MAE: 266.53 units).

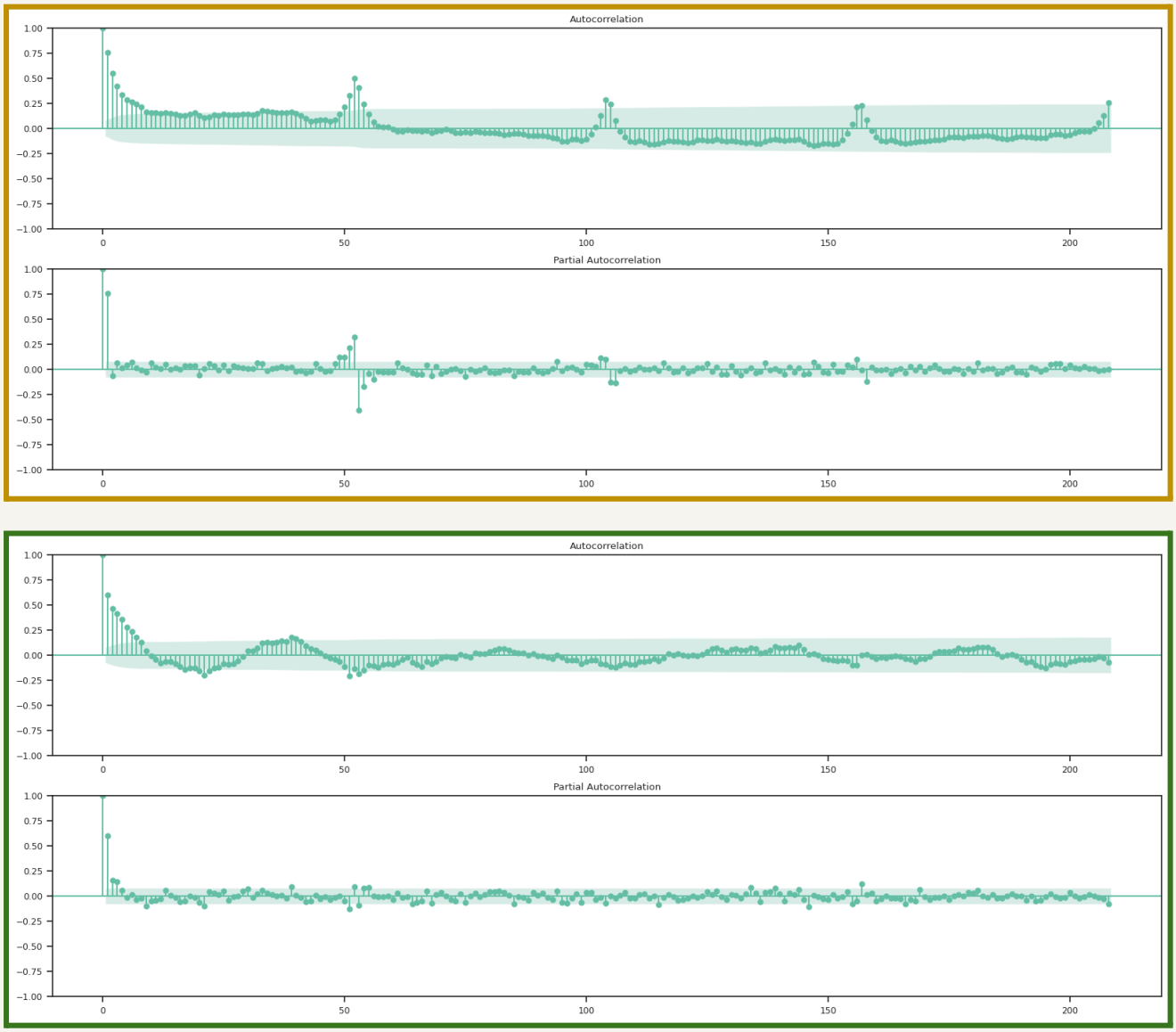


Figure S2. Autocorrelation (ACF) and partial autocorrelation (PACF) of weekly sales for The Alchemist, showing strong short-term dependence and clear annual seasonality. *Top*: Before decomposition, the ACF decays slowly, indicating non-stationarity, with strong seasonal spikes at Lag 52 and its multiples. The PACF is dominated by a strong lag-1 spike, with a secondary significant spike at lag 52, while most other lags fall within confidence bounds. These patterns indicate strong week-to-week dependence with annual seasonality. *Bottom*: After STL decomposition, these seasonal spikes are not present in the residuals and the ACF decays faster, showing that trend and seasonality have been removed. However, Lag 1 remains significant, and the Ljung-Box test indicates the residuals are not white noise, meaning some stochastic structure remains in the residuals beyond what STL can capture.

The Very Hungry Caterpillar

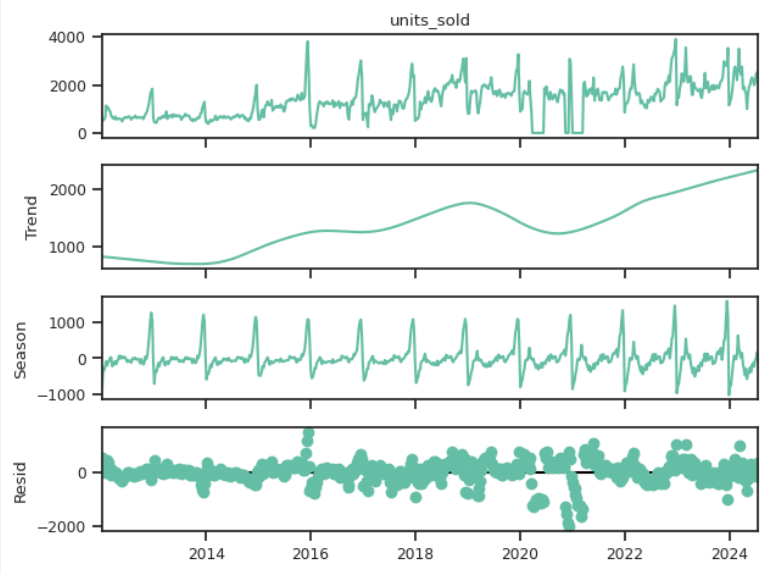


Figure S3. STL decomposition of weekly sales for The Very Hungry Caterpillar (trend=157, seasonal=25, period=52). Seasonal-Trend decomposition showing strong trend strength (0.59) with persistent upward growth and noticeable COVID-19 disruption (2020-2021), moderate seasonal component (0.45) with consistent annual cycles, and residuals displaying irregular high-amplitude spikes and extreme COVID-period drop (MAE: 266.53 units). The dominant trend component drives the series dynamics, with seasonality providing secondary but clear annual patterns superimposed on the growth trajectory.

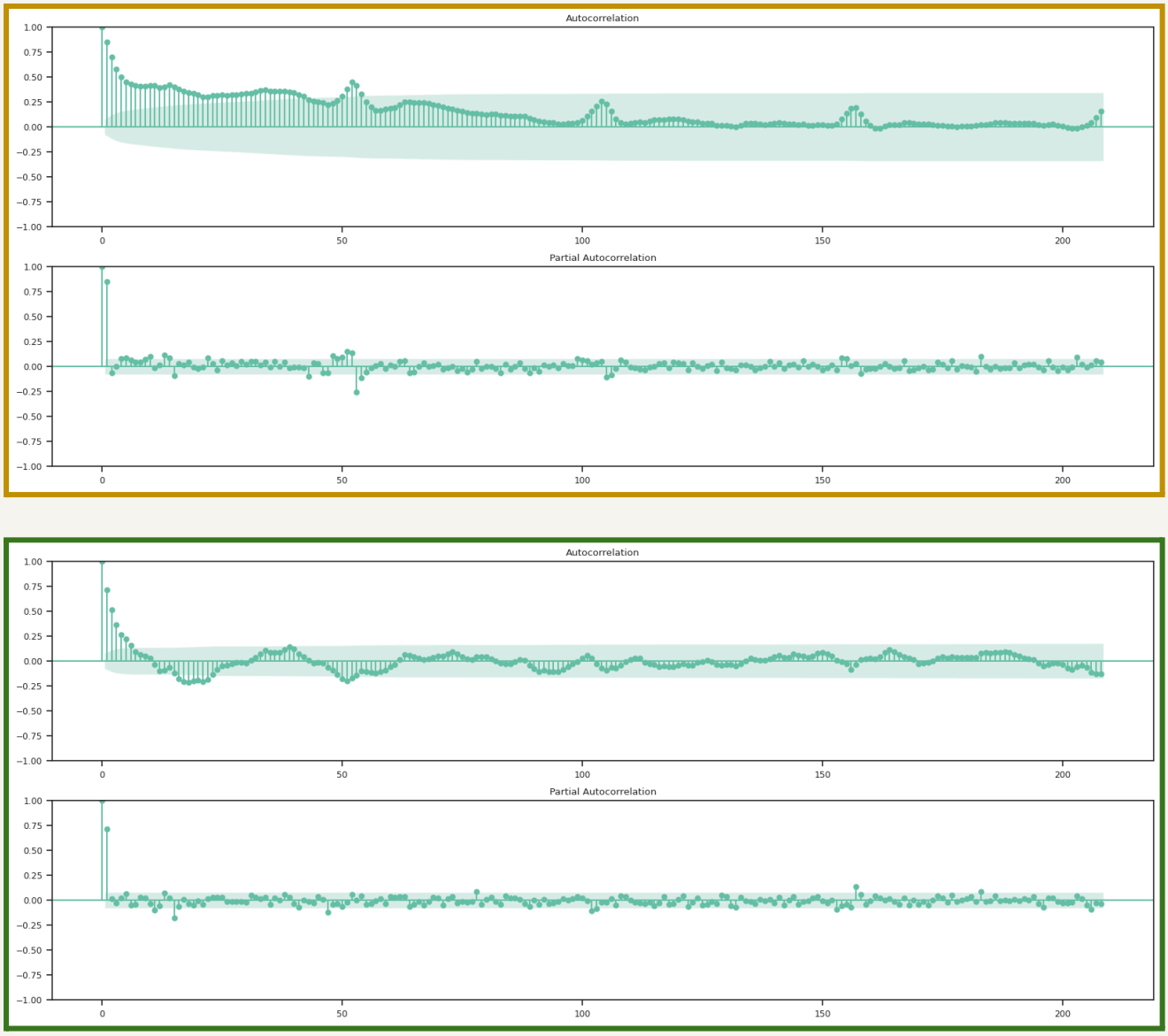


Figure S4. Autocorrelation (ACF) and partial autocorrelation (PACF) of weekly sales for The Very Hungry Caterpillar, showing strong trend and clear annual seasonality. *Top*: Before decomposition, the ACF exhibits very strong persistence with slow exponential decay, remaining above confidence bounds for about 50 lags, and clear spikes at lags 52 and its multiples, indicating annual seasonality. The PACF is dominated by an extremely strong lag-1 effect, followed by a rapid cutoff, with only a minor seasonal spike at lag 52. This pattern indicates a dominant underlying trend with superimposed annual cycles. Week-to-week dynamics are less strong than the trend component. *Bottom*: After STL decomposition, these seasonal spikes are less present in the residuals and the ACF decays faster, showing that trend and seasonality have been removed to some extent. However, Lag 1 remains significant, and the Ljung-Box test indicates the residuals are not white noise, meaning some stochastic structure remains in the residuals beyond what STL can capture.

Monthly Aggregated Data

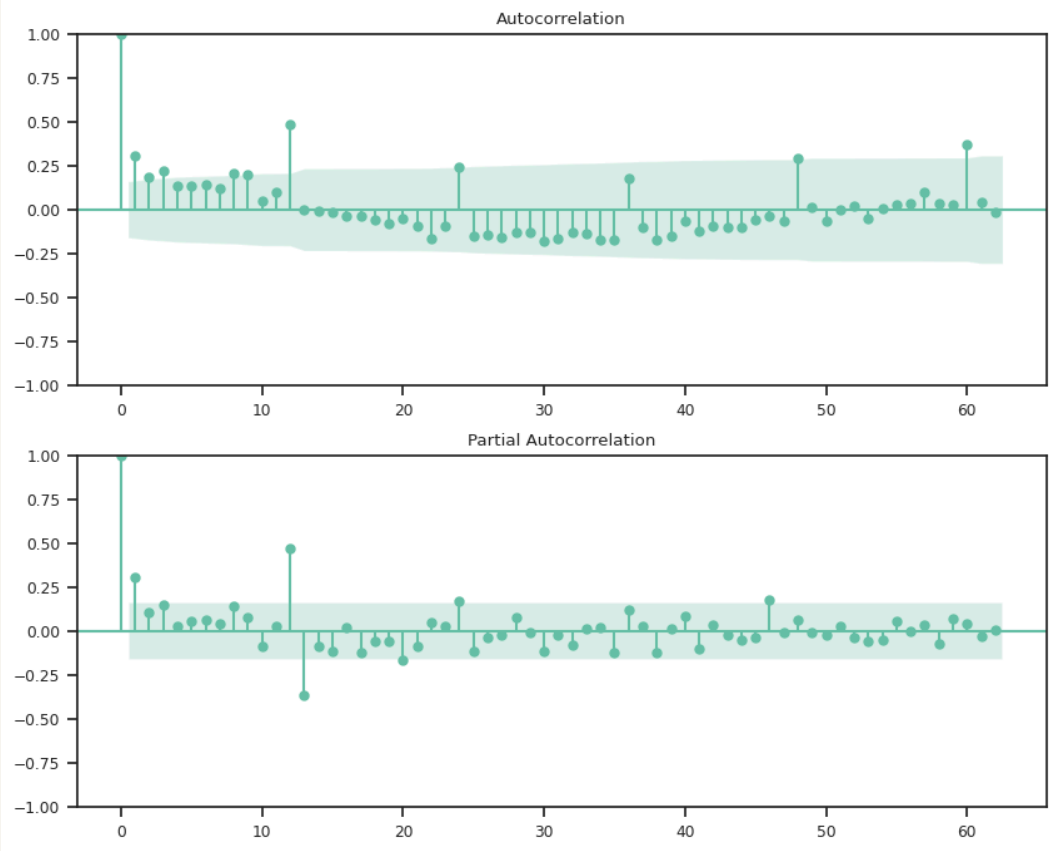


Figure S5. Autocorrelation (ACF) and partial autocorrelation (PACF) of monthly aggregated sales for The Alchemist, highlighting clear annual seasonality. The ACF shows moderate short-term dependence with rapid decay after lag 1, alongside a significant spike at lag 12, indicating strong annual seasonality. The PACF exhibits significant effects at lag 1 and lag 12, with most remaining lags within confidence bounds, suggesting that monthly sales are mainly driven by the previous month and the same month of the prior year.

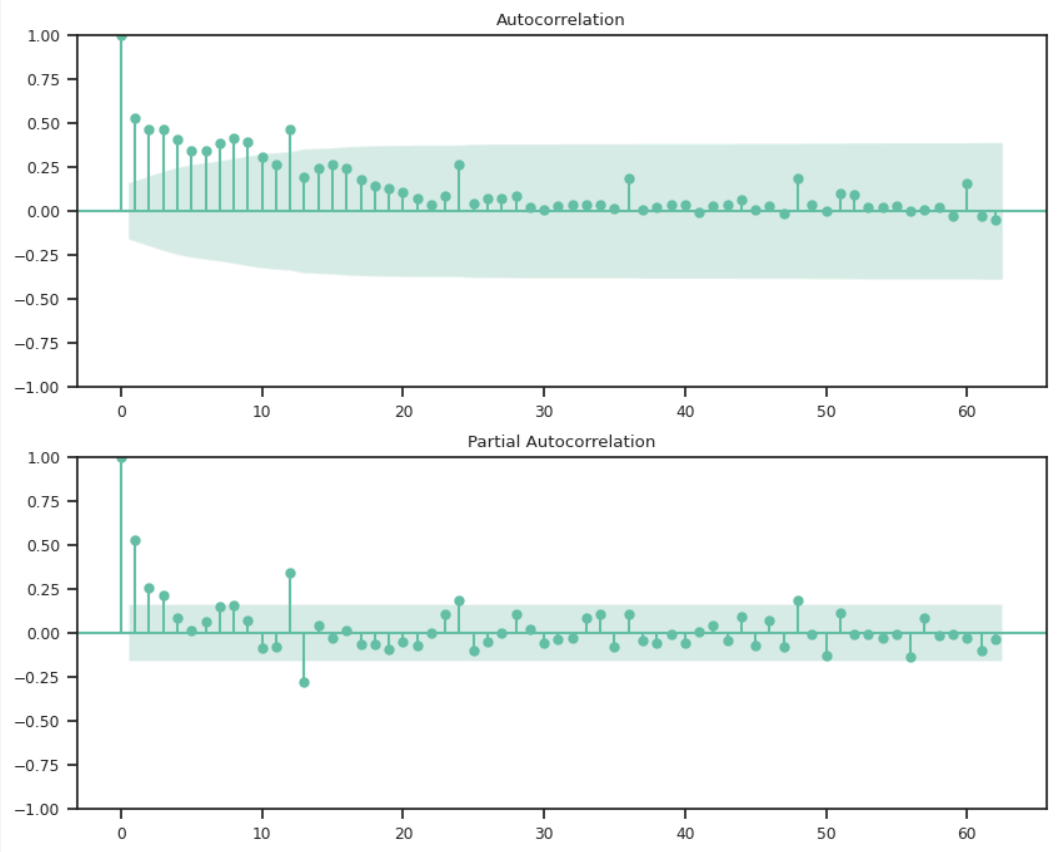


Figure S6. Autocorrelation (ACF) and partial autocorrelation (PACF) of monthly aggregated sales for The Very Hungry Caterpillar. The ACF shows strong persistence with slow decay and a clear spike at lag 12, indicating an underlying trend with annual seasonality. The PACF is dominated by lag 1, with smaller contributions from recent months and a strong lag-12 effect. Together, these patterns show that monthly aggregated sales are driven by a trend component and a strong annual cycle, which monthly aggregation preserves while smoothing out weekly volatility.

LSTM

LSTM underperformed throughout, but this reflects the limits of the available data rather than the limits of the approach. 623 weekly observations is a modest sample for deep learning, and with more data, explicit treatment of COVID-period anomalies, sample weighting to prioritise high-value peak weeks, or attention-based architectures such as Transformers, neural models may yet outperform classical methods.

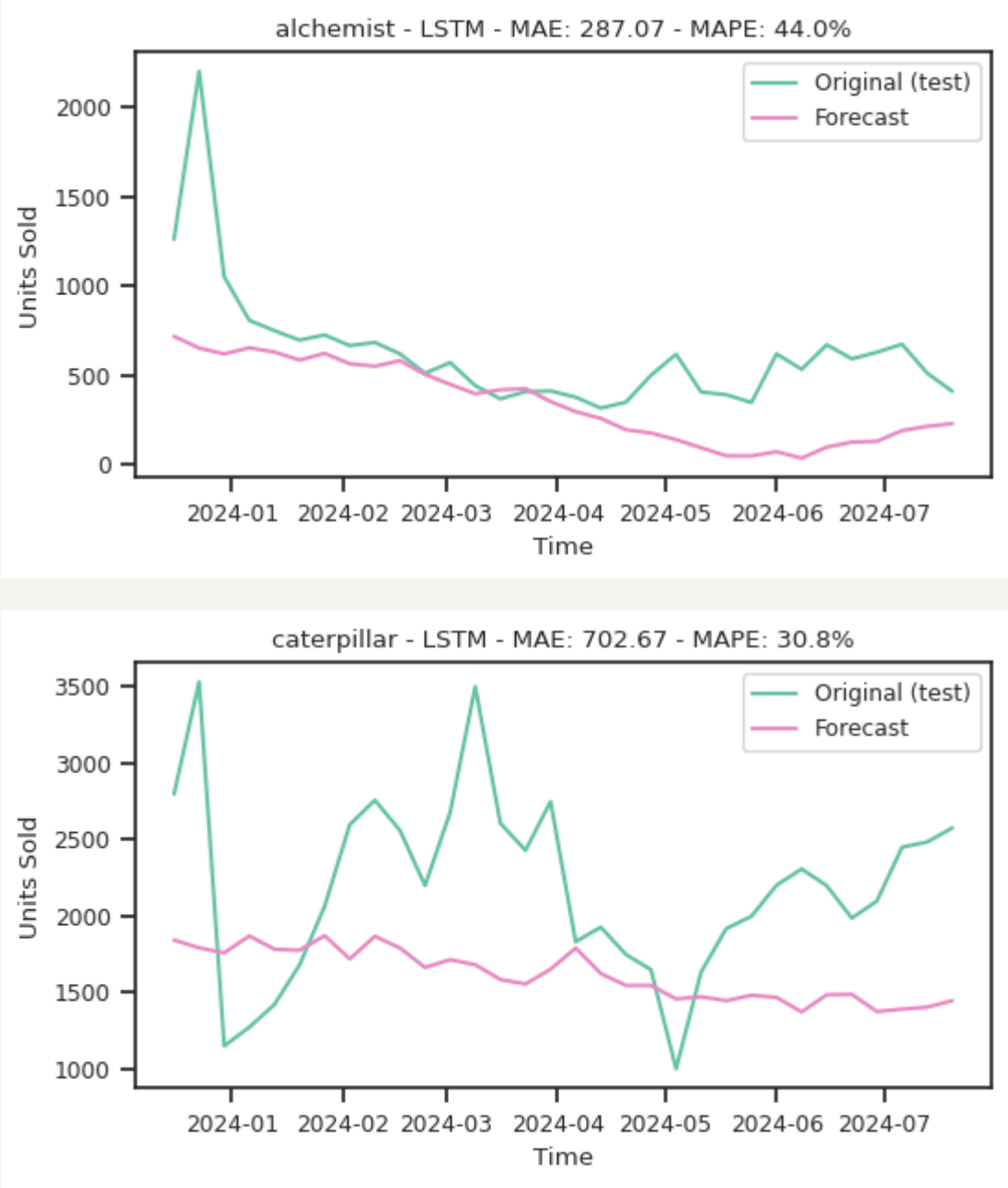
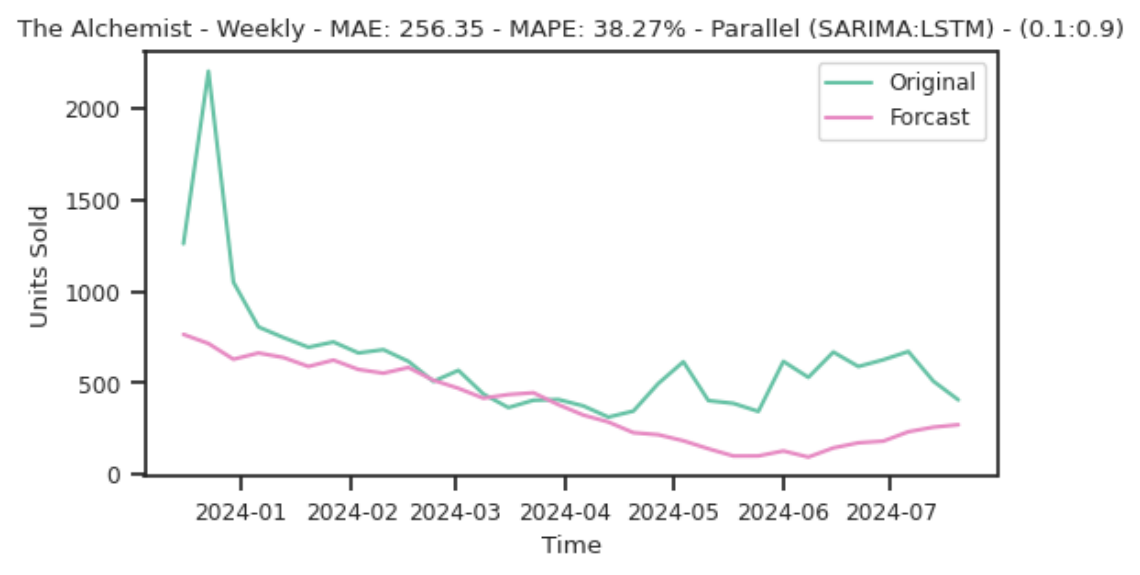


Figure S7. LSTM forecast performance for The Alchemist (*top*) and The Very Hungry Caterpillar (*bottom*) (weekly data, 32-week horizon). The Alchemist shows conservative predictions that systematically underestimate the initial spike and subsequent volatility (MAPE: 44.0%). The Very Hungry Caterpillar demonstrates similar conservative behaviour, (MAPE: 30.8%). Both models fail to capture peak magnitudes and timing, likely due to limited training data, and COVID-period contamination.

Weighted Ensemble Modelling (SARIMA + LSTM)



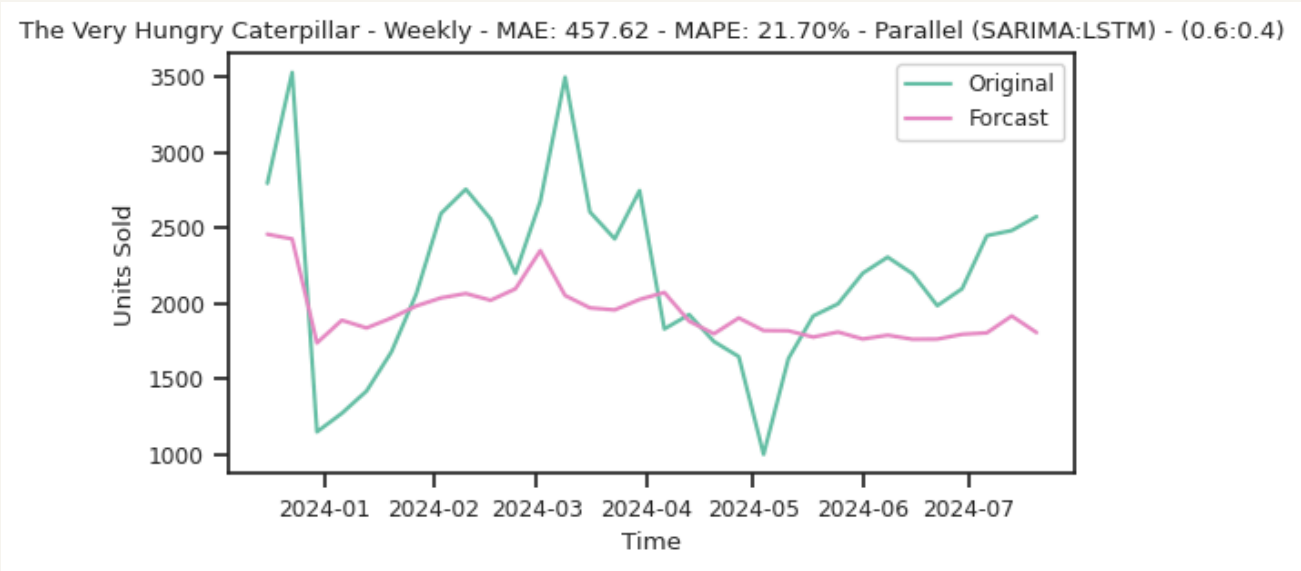


Figure S8. Weighted ensemble forecasts (SARIMA + LSTM) for The Alchemist (*top*) and The Very Hungry Caterpillar (*bottom*) (weekly data, 32-week horizon). Parallel ensemble forecasts combining SARIMA and LSTM predictions through weighted averaging. The Alchemist achieved optimal performance with 10% SARIMA and 90% LSTM weighting (MAPE=38.27%, MAE=256.35). The Very Hungry Caterpillar used 60% SARIMA and 40% LSTM weighting (MAPE=21.70%, MAE=457.62). Both ensembles show smoother, more conservative forecasts than actual volatility, demonstrating that weighted averaging cannot overcome the fundamental limitation of LSTM's inability to capture peak dynamics with limited training data. The results suggest ensemble methods provide minimal benefit when one component model significantly underperforms.

Log transformation - The Very Hungry Caterpillar

For The Very Hungry Caterpillar, we tested two extensions to the baseline SARIMA model: log transformation to stabilise variance and a binary COVID exogenous variable to explicitly model the pandemic's structural break. However, empirical evaluation showed neither extension meaningfully improved forecast accuracy. Log transformation caused model instability and poor performance (Fig. S11), while the COVID indicator did not significantly reduce errors. Therefore, we retained the simpler SARIMA specification trained on raw, untransformed data, as it achieved the best balance of accuracy and interpretability.

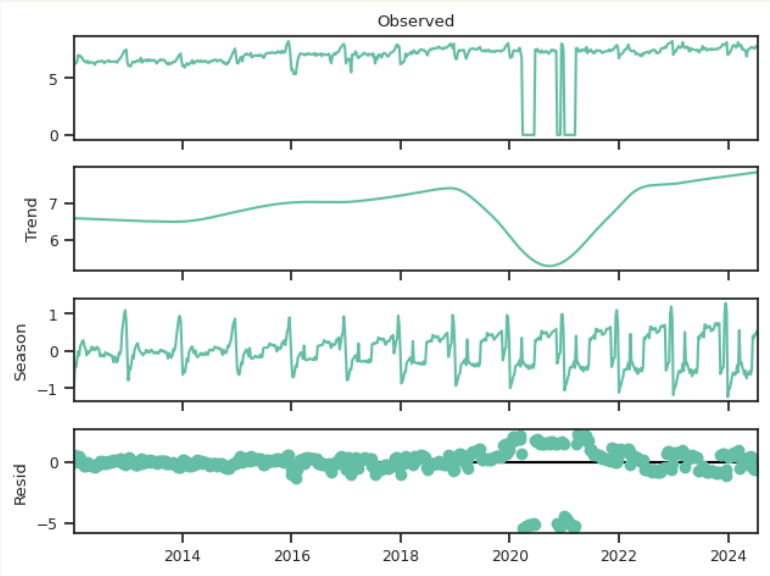


Figure S9. STL decomposition of Log-transformed weekly sales for The Very Hungry Caterpillar (trend=157, seasonal=25, period=52). Seasonal-Trend decomposition of log-transformed data showing strong upward trend with severe COVID-19 disruption, consistent seasonal cycles, and residuals with reduced heteroskedasticity but large negative COVID-period deviations (MAE: 597.23 units after back-transformation).

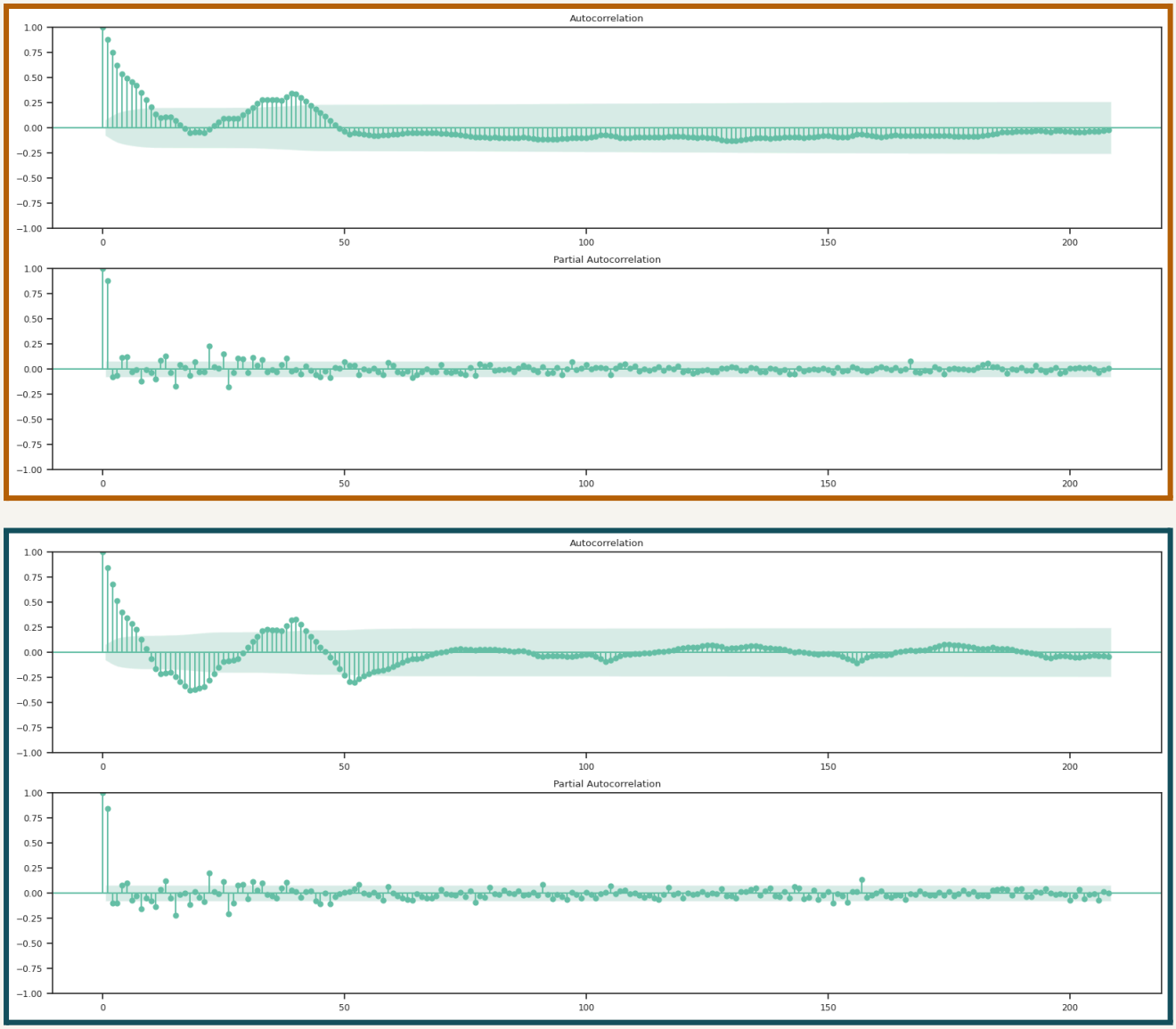
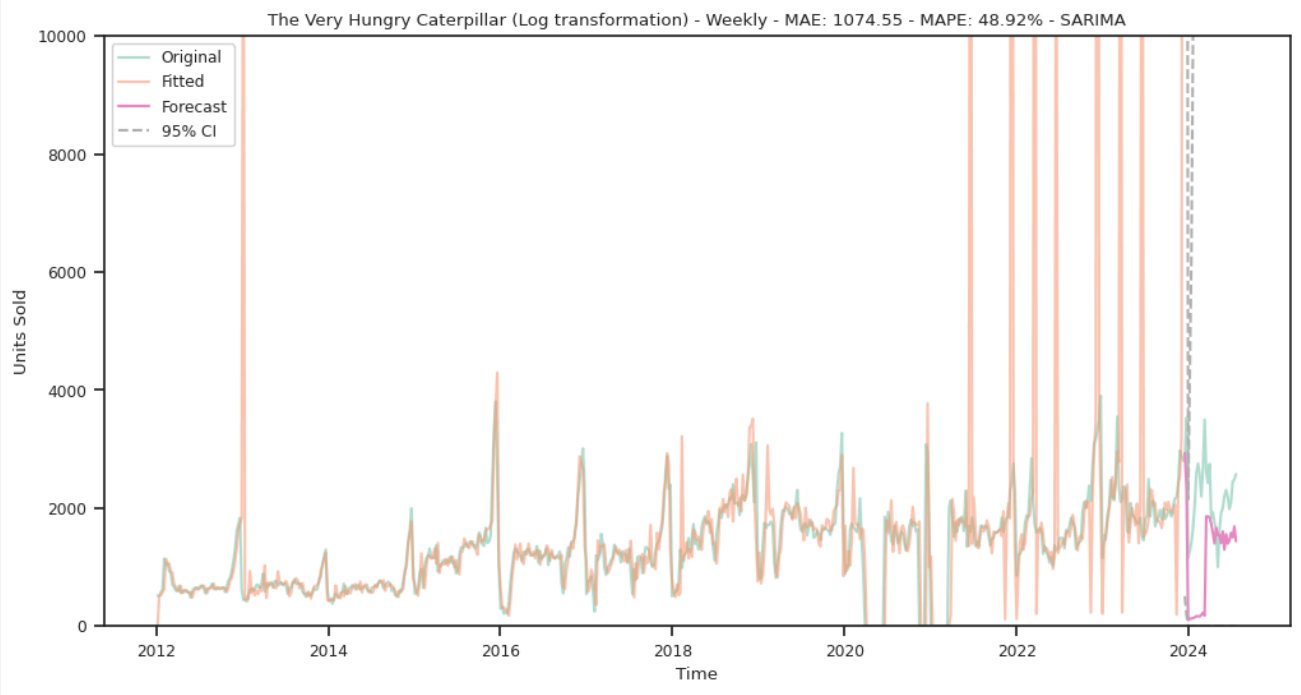


Figure S10. Autocorrelation (ACF) and partial autocorrelation (PACF) of log-transformed weekly sales for The Very Hungry Caterpillar. *Top*: Before decomposition, the ACF shows rapid initial decay, strong short-term persistence with clear seasonal structure, while the PACF is dominated by a strong lag-1 effect, indicating that the series still contains trend, despite variance stabilisation. *Bottom*: After STL decomposition, the seasonality is reduced but the ACF still exhibits multiple statistically significant lags beyond the confidence bounds, indicating substantial remaining autocorrelation. Persistent significant autocorrelation and a non-white-noise Ljung-Box result indicate remaining stochastic structure is not fully captured by decomposition alone.



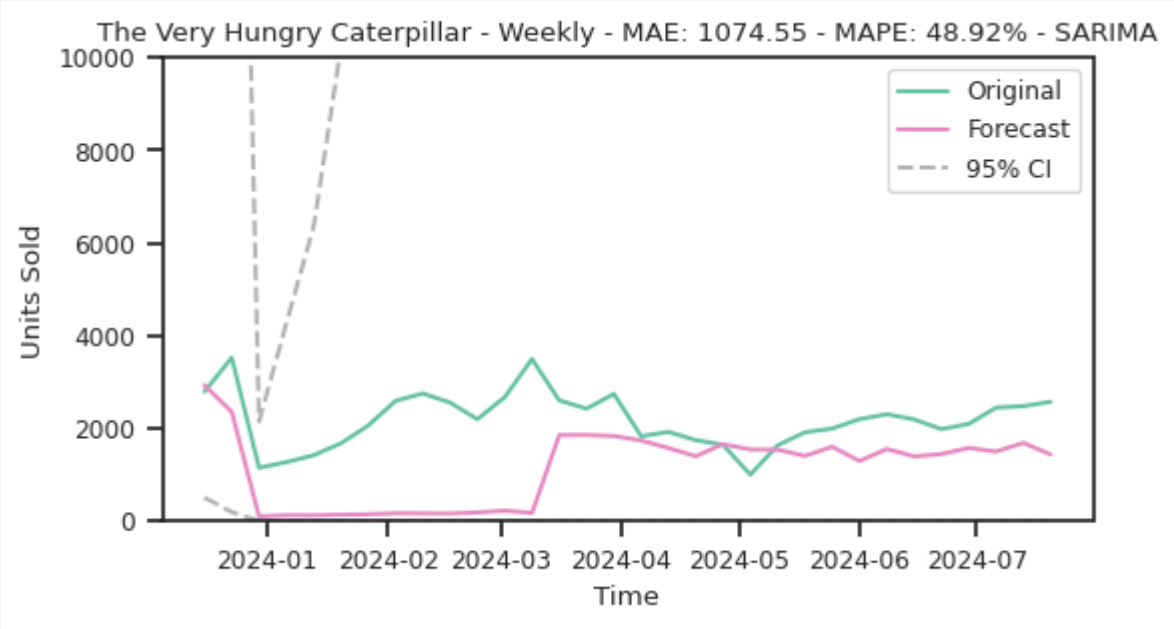


Figure S11. SARIMA forecast performance on Log-transformed weekly sales for The Very Hungry Caterpillar. *Top*: Full time series showing fitted values closely tracking original data with extreme spikes during 2022-2024, and exploding confidence interval expansion. *Bottom*: The model predicts unrealistic near-zero values in early 2024 before recovering to baseline (MAPE=48.92%, MAE=1074.55). The exploding confidence intervals indicate severe numerical instability caused by back-transforming log-space predictions of near-zero COVID-period values. This instability, combined with the model's failure to capture trend and seasonal patterns in the forecast period, demonstrates why log transformation was rejected in favour of raw-data modeling despite its theoretical success in variance stabilisation.

Parameters and Settings

STL Decomposition		
Parameter	Value	Notes
period	52	weekly data with annual seasonality
seasonal windows	[7, 13, 25]	testing adaptive to consistent seasonal patterns
trend windows	[79, 157, 261]	roughly 1.5, 3, and 5 times the period value
robust	False	True underestimates seasonal strength
alpha	0.05	significance level

Table S2. STL decomposition parameters and settings for weekly units-sold time series.

Weekly Data	Parameter	Value		Notes
SARIMAX (auto_arima)	forecast_horizon	32		Prediction horizon (weeks)
	transformation	False		No variance stabilisation
	exogenous	False		No special adjustments were applied to the COVID-19 anomaly interval;
	seasonal	True		SARIMA enabled
	m	52		Annual seasonality (weeks)
	start_p	1		
	d	The Alchemist	None	
		The Very Hungry Caterpillar	1	To ensure stationarity
	start_q	1		
	max_p	3		
	max_d	2		
	max_q	3		
	star_P	1		
	D	None		
	start_Q	1		
	max_P	2		
	max_D	1		
	max_Q	2		
	max_order	The Alchemist	5	Prevents overfitting
		The Very Hungry Caterpillar	8	Allows more complex model
	stepwise	True		Stepwise search for efficiency
	information_criterion	aic		
	stationarity	False		
	scoring	mse		
	maxiter	100		

Table S3. SARIMA model configurations and implementation details for weekly units-sold time-series forecasting.

	Parameter	Value		Notes
XGBoost (Weekly Data)	forecast_horizon	32		Prediction horizon (weeks)
	model	The Alchemist	additive	Trend type in model
		The Very Hungry Caterpillar	multiplicative	
Regressor	n_estimators	200		Avoids overfitting
	learning_rate	0.05		Conservative learning rate
	booster	gbtree		Gradient boosted trees
	tree_method	exact		Standard tree construction
	objective	reg:squarederror		Standard loss function for regression
	max_depth	5		Limits tree depth
	min_child_weight	3		Prevents overfitting
	gamma	0.1		Minimum loss reduction
	reg_alpha	0		For time series, consecutive lags matter
	reg_lambda	1		L2 regularization
	subsample	0.8		Fraction of data per tree
	random_state	42		Reproducibility
	base_score	0.5		Initial prediction score
Predictor	sp (seasonality)	52		Annual seasonality (weeks)
	degree	The Alchemist	1	Relatively flat trend; Linear is sufficient
		The Very Hungry Caterpillar	2	Upwards trend and accelerating growth; Quadratic is appropriate
	window_length	The Alchemist	52	Lookback window for features
		The Very Hungry Caterpillar	104	
	model type	The Alchemist	additive	Trend type in model
		The Very Hungry Caterpillar	additive or multiplicative	
	strategy	recursive		Forecast 1-step ahead recursively
Grid Search	initial_window	0.7		Fraction of training data for CV
	strategy	refit		Refit on full data after CV
	scoring	MAE		Mean Absolute Error
	window_length	The Alchemist	[52, 78, 104]	A bias-variance trade-off: Shorter windows capture recent patterns, higher variance; Longer windows capture more stable, but older patterns. Avoid including Covid-19 period.
		The Very Hungry Caterpillar	[52, 78, 104, 130]	

Table S4. XGBoost model configurations and implementation details for weekly units-sold time-series forecasting.

	Parameter	Value	Notes
LSTM (Weekly Data)	lookback	104	Required history
	forecast_horizon	32	Multi-step ahead forecasting (weeks)
	validation_split	0.20	Hold-out validation fraction
Network Architecture	hidden_activation	tanh	Standard LSTM activation
	output_activation	linear	Regression output
	dropout_rate	0.20	Regularisation to reduce overfitting
Optimisation	loss	mse	Mean Squared Error
	metrics	mae	Interpretable error metric
Training	epochs	100	Upper bound on training iterations
	batch_size	32	Larger values increase the dominance of long-term patterns; Emphasis on short-term patterns to detect the rare peaks
	early_stopping_ monitor	val_loss	Prevents overfitting
	patience	10	Early stopping tolerance
Checkpoints Monitors	loss	val_loss (min)	Save best model by validation loss
	mae	mae (min)	Secondary performance criterion
Hyperparameter Tuning	n_neuron	[32, 16]	conservative - less overfitting - avoids learning noise
		[64, 32]	balanced
		[128, 64, 32]	More complex - more capacity - risk of overfitting
	optimizer	[adam, rmsprop]	Alternative optimisation strategies
	learning_rate	[0.0005, 0.001]	Stability vs convergence trade-off
	tuner_objective	val_loss	Consistent with training objective

Table S5. LSTM model configurations and implementation details for monthly units-sold time-series forecasting.

	Parameter	Value	Notes
Hybrid (Sequential)	lookback	104	Required history
	forecast_horizon	32	Multi-step ahead forecasting (weeks)
	validation_split	0.20	Hold-out validation fraction
Network Architecture	hidden_activation	tanh	Standard LSTM activation
	output_activation	linear	Regression output
	dropout_rate	0.20	Regularisation to reduce overfitting
Optimisation	loss	mse	Mean Squared Error
	metrics	mae	Interpretable error metric
Training	epochs	50	Less training iterations for forecasting residuals
	batch_size	32	Emphasises short-term patterns to detect the rare peaks
	early_stopping_ monitor	val_loss	Prevents overfitting
	patience	7	More aggressive than LSTM-only analysis
Checkpoints Monitors	loss	val_loss (min)	Save best model by validation loss
	mae	mae (min)	Secondary performance criterion
Hyperparameter Tuning	n_neuron	[8, 8]	residuals exhibit simpler, more stationary patterns after SARIMA removes trend and seasonality
		[16, 8]	
		[32, 16]	
	optimizer	[adam, rmsprop]	Alternative optimisation strategies
	learning_rate	[0.0005, 0.001]	Stability vs convergence trade-off
	tuner_objective	val_loss	Consistent with training objective

Table S6. LSTM model configurations and implementation details for weekly units-sold time-series forecasting of SARIMA residuals (sequential hybrid modelling).

Monthly Data	Parameter	Value		Notes
SARIMAX (auto_arima)	forecast_horizon	8		Prediction horizon (weeks)
	transformation	False		No variance stabilisation
	exogenous	False		No special adjustments were applied to the COVID-19 anomaly interval;
	seasonal	True		SARIMA enabled
	m	12		Annual seasonality (months)
	start_p	1		
	d	1		To ensure stationarity
	start_q	1		
	max_p	3		
	max_d	2		
	max_q	3		
	star_P	1		
	D	1		To ensure stationarity
	start_Q	1		
	max_P	2		
	max_D	1		
	max_Q	2		
	max_order	The Alchemist	5	Prevents overfitting
		The Very Hungry Caterpillar	8	Allows more complex model
	stepwise	True		Stepwise search for efficiency
	information_criterion	aic		
	stationarity	False		
	scoring	mse		
	maxiter	50		Less iteration than weekly modelling

Table S7. SARIMA model configurations and implementation details for monthly units-sold time-series forecasting.

	Parameter	Value		Notes
XGBoost (Monthly Data)	forecast_horizon	8		Prediction horizon (months)
	model	The Alchemist	additive	Trend type in model
		The Very Hungry Caterpillar	additive or multiplicative	
Regressor	n_estimators	200		Avoids overfitting
	learning_rate	0.05		Conservative learning rate
	booster	gbtree		Gradient boosted trees
	tree_method	exact		Standard tree construction
	objective	reg:squarederror		Standard loss function for regression
	max_depth	5		Limits tree depth
	min_child_weight	3		Prevents overfitting
	gamma	0.1		Minimum loss reduction
	reg_alpha	0		For time series, consecutive lags matter
	reg_lambda	1		L2 regularization
	subsample	0.9		Fraction of data per tree; Less aggressive for monthly data;
	random_state	42		Reproducibility
	base_score	0.5		Initial prediction score
Predictor	sp (seasonality)	12		Annual seasonality (months)
	degree	The Alchemist	1	Relatively flat trend; Linear is sufficient
		The Very Hungry Caterpillar	2	Upwards trend and accelerating growth; Quadratic is appropriate
	window_length	The Alchemist	12	Lookback window for features
		The Very Hungry Caterpillar	24	
	model type	The Alchemist	additive	Trend type in model
		The Very Hungry Caterpillar	additive or multiplicative	
	strategy	recursive		Forecast 1-step ahead recursively
Grid Search	initial_window	0.7		Lower fraction of training monthly data for CV
	strategy	refit		Refit on full data after CV
	scoring	MAE		Mean Absolute Error
	window_length	The Alchemist	[12, 18, 24]	A bias-variance trade-off: Shorter windows capture recent patterns, higher variance; Longer windows capture more stable, but older patterns. Avoid including Covid-19 period.
		The Very Hungry Caterpillar	[12, 18, 24, 36]	

Table S8. XGBoost model configurations and implementation details for monthly units-sold time-series forecasting.